Supplementary Table 1: Characteristics of Included Studies

Study Author and Date	Country of Subjects	Diagnosis of interest	eRAT form	Nomenclature used in study	Research Participant	eRAT exposure	Data Collection Methodology	Data Analysis	Risk/uncertainty theme in 1° analysis	CASP Quality Assessment
Babb de Villiers 2019	UK and international	Cancer	Prognostic	Risk prediction models	Clinicians	Workshop	Workshop	Thematic Analysis	No	**2**
Chiang 2015	Australia	Cancer	Clinical	Risk tool	Clinicians	Simulated Clinic	Semi-structured Interviews	NPT	No	**8**
Coe 2016	USA	Breast Cancer	Prognostic	Risk Assessment	Patients	Workshop/Usability testing	"Morae" software recording.	Framework Analysis (implied not stated)	No	**8**
Collins 2013	Australia	Breast cancer	Prognostic	Risk assessment and decision support	Clinicians	Workshop	Focus groups	Unclear	No	**4**
Colombet 2003	France	Cardiovascular Disease, Cancer, MH risks	Prognostic	Personalised Estimate of Risk	Clinicians	Workshop/Usability testing	Focus group	Framework Analysis (implied not stated)	Yes	**6**
Crawford 2010	UK	DM foot disease	Prognostic	Risk Assessment Tool	Clinicians	Real World Use	Semi-structured interviews	Thematic Analysis	No	**8**
Dikimitis 2015	UK	Cancer	Clinical	Electronic Risk Assessment Tools	Clinicians	Real World Use	Semi-structured interviews	NPT	No	**8**
Du 2010	China	Cardiovascular Disease	Prognostic	Risk Assessment Tools	Clinicians	Workshop/Usability testing	Semi Structured focus groups	Inductive coding	No	**3**

Eaton 2009	USA	Cardiovascular Disease	Prognostic	Interactive Risk Communication and Decision Support	Patients	Real World Use	Focus Groups	Unclear	Yes	**4**
Harry 2019	USA	Cancer Cardiovascular Disease	Prognostic	Clinical Decision Support	Clinicians	Real World Use	Semi-structured interviews	Grounded Theory	No	**9**
Hawking 2019	UK	Cardiovascular Disease	Prognostic	Risk Report	Patients	Real World Use	Semi-structured interviews	Thematic Analysis	Yes	**8**
Jiwa 2006	USA	Cancer	Clinical	Interactive referral proforma	Clinicians	Real World Use	Semi structured interviews	Thematic Analysis	No	**7**
Keogh 2019	Australia and New Zealand	Breast Cancer	Prognostic	Risk assessment and management support tool	Clinicians and Patients	Workshop/Usability Testing	Focus Group	Thematic Analysis (implied not stated)	No	**8**
Kidney 2017	UK	Colorectal Cancer	Clinical	Information based local evaluation	Clinicians and Practice Managers	Real World Use	Interviews	Thematic Analysis	Yes	**6**
Kukafka 2015	USA	Breast Cancer	Prognostic	Decision aid	Patients	Real World Use	Focus groups	Thematic Analysis (implied not stated)	Yes	**8**
McParland 2019	UK	DDx	DDx	Decision Support System	Clinicians	Real World Use	Focus groups	Thematic Analysis	No	**8**
Nieroda 2018	UK	Cancer	Clinical	Decision Support Tool	Clinicians and Patients	Workshop/Usability Testing	Focus groups, think aloud use of website, open	Thematic Analysis	No	**8**

							ended questionnaires			
Peiris 2009	Australia	CVD	Prognostic	Clinical Decision Support	Clinicians	Real World Use	Semi-structured interviews	Thematic Analysis	Yes	**8**
Porat 2017	UK	DDx	DDx	Decision Support System	Clinicians	Simulated Clinic	Semi-structured interviews	Thematic Analysis	No	**7**
Porat 2018	UK	Stroke	Prognostic	Decision Aid	Clinicians	Workshop/Usability Testing	Semi-structured interviews	Thematic Analysis	Yes	**3**
Porat 2019	UK	Stroke	Prognostic	Decision Aid	Clinicians, Patients, and other Stakeholders	Workshop/Usability testing	Focus groups, Interviews and usability evaluations.	Thematic Analysis	Yes	**5**
Richardson 2019	USA	Pharyngitis and Cough	Clinical	Decision Support Tools	Clinicians	Real World Use	Interview, video and audio of consultation	Constant Comparative Method	No	**8**
Saver 2014	USA	Diabetes complications	Prognostic	Outcome Model	Patients	Real World Use	Interviews	general inductive approach	Yes	**5**
Short 2003	UK	Stroke	Prognostic	Decision Support System	Clinicians	Simulated Clinic	Qualitative interview and questionnaire	Unclear	No	**7**
Short 2004	UK	Stroke	Prognostic	Decision Support System	Clinicians	Workshop/Usability testing	Interviews	Constant Comparative Method	Yes	**8**
Vail 2012	UK	Mental Health risk	Prognostic	Decision Support System	Clinicians	Workshop/Usability testing	Interviews, focus groups	Thematic Analysis	No	**8**

Voruganti 2015	Canada	Multiple	Prognostic	Risk Assessment Tools	Clinicians	Workshop/Usability testing	Focus Groups and usability testing	Grounded theory	No	**8**
Walker 2017	Australia	Colorectal Cancer	Prognostic	Risk Prediction Tool	Clinicians and Practice Managers	Simulated Clinic	Interviews, videoed simulated consultations analysed	Framework Analysis	No	**7**

Supplementary Table 2: Assessment of quality of studies with the CASP tool

Study ID	1. Was there a clear statement of the aims of the research?	2. Is a qualitative methodology appropriate?	3. Was the research design appropriate to address the aims of the research?	4. Was the recruitment strategy appropriate to the aims of the research?	5. Was the data collected in a way that addressed the research issue?	6. Has the relationship between researcher and participants been adequately considered?	7. Have ethical issues been taken into consideration?	8. Was the data analysis sufficiently rigorous?	9. Is there a clear statement of findings?	10. How valuable is the research?
Babb de Villiers 2019	Yes	Yes	No	No	Can't tell	No	No	No	No	No
Chiang 2015	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Coe 2016	Yes	Yes	Can't tell	Yes	Yes	No	Yes	Yes	Yes	Yes
Collins 2013	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Yes	Yes
Colombet 2003	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Crawford 2010	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Dikimitis 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Du 2010	Yes	Yes	Yes	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell	Can't tell
Eaton 2009	Yes	Yes	Can't tell	Can't tell	Yes	No	No	Can't tell	Yes	Yes
Harry 2019	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Hawking 2019	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Jiwa 2006	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Can't tell	Can't tell
Keogh 2019	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Kidney 2017	Yes	Yes	Yes	Can't tell	Yes	No	Yes	Can't tell	Yes	Yes
Kukafka 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
McParland 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Nieroda 2018	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Peiris 2009	Yes	Yes	Yes	Yes	Yes	Can't tell	Yes	Yes	Yes	Yes
Porat 2017	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Porat 2018	No	Yes	Can't tell	Yes	Yes	No	No	No	Yes	Can't tell
Porat 2019	Yes	Yes	Yes	Yes	Yes	No	Can't tell	Can't tell	Yes	Can't tell
Richardson 2019	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Saver 2014	Yes	Yes	Can't tell	Yes	Can't tell	No	Yes	Yes	Yes	Can't tell
Short 2003	Yes	Yes	Yes	Can't tell	Yes	Can't tell	Yes	Yes	Yes	Yes
Short 2004	Yes	Yes	Can't tell	Yes	Yes	No	No	No	Yes	Yes
Vail 2012	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Voruganti 2015	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Walker 2017	Yes	Yes	Yes	Yes	Can't tell	Can't tell	Yes	Yes	Yes	Yes

Supplementary Table 3: Illustrative Quotes to reinforce Analytical Themes

Theme		Quote (all clinician comments, unless stated)	Study

Novel Risk	*Alerts to new risk*	"*It's just electronic highlight, makes you think even if you immediately dismiss it, at least that millisecond you've thought about it, and I think that is going to be useful at some point, but for how many people I don't know*"	Dikomitis 2015
		"Helpful to refresh your memory about things that are rare…we mainly stick to the phrase: 'common things are common"	Porat 2017
	Alerts to alternative risks	*"Yes, I must admit ovarian didn't come so high up, I had already fixated on the bowel stuff, which maybe I shouldn't have. This really said hey, consider ovarian as well. So again that was useful, yes"*	Chiang 2015
		"I think it widens your diagnosis, like the guy that had COPD and then ended up with AS. I would get to the diagnosis, but it would be longer without the tool."	Porat 2017
	Alerts to risk factor	*"the patient can see very clearly, [and say] 'Wow, smoking contributes a lot to cardiovascular risk, I do actually want to think about smoking cessation'"*	Voruganti 2015
		"I don't think it's (the eRAT) altered practice. I think it's kept some of the not so hard red flag symptoms in people's minds…"	Kidney 2017
Risk Refinement	*Orientation of risk (within guidelines/clinical pathways)*	*"Normally I'd get a few investigations, get the results back and then based on that say do we need to do something, or I refer this on based on that. But I guess if I have a calculator saying it's higher risk, it might prompt me to make a referral to a specialist a bit earlier."*	Chiang 2015
		"Think the biggest use I got, because you presented with very constitutional symptoms that a lot of things could present with, this was useful for me to think what's the good next step and what should I think of. So I was thinking in my mind more"	Chiang 2015
	More accurate risk, reduction of uncertainty	*For clinicians, the main perceived benefit was iPrevent's provision of accurate and reliable risk estimates for both ten year and lifetime risk, and standardized risk management advice for women*	Keogh 2019, Researcher comment
		"It is helpful in that it reminds me that the risk of stroke goes up substantially as you get older so your benefit (of aspirin) increases"	Short 2003
	Simplification of risk, reduction of uncertainty	*"Well I suppose you could think about the common comorbidities, so hypertension and stroke, AF (atrial fibrillation) and stroke, diabetes and stroke and you could think about not necessarily an algorithm but a sort of stepwise prioritisation about what you should think about in terms of the patient's total management, you know, which would be the most important area of focus?"*	Porat 2019

	Introduction of new risk knowledge	*"Other colleagues have said as well we're far more aware of thrombocytosis and increased platelets. We weren't aware that tended to increase the score for increased cancer risk"*	Dikomitis 2015
		"The learning aspect is very important, since this system is based on evidence and evidence can change"	Porat 2019
Communication	*Patient to Clinician Communication*	*"I think it's important to be informed and to get there as early as possible. To go and see the GP, and if there's a tool perhaps, you know, to help me do this, that may be of interest"*	Nieroda 2018 Patient Comment
	Risk focuses conversations	*"it usually stops a lot of the meandering dialogue that you'd otherwise engage in…. [initiate] that discussion in terms of what their risk factors are*"	Voruganti 2015
		"So having that piece of paper there gave you that conversation: 'Well now we've finished everything, let's look at this.'"	Peiris 2009
	Risk justifies action	*"But your heart is beating kinda fast, you've had a fever last night...the recommendation would be to get a chest x-ray today."*	Richardson 2019
		"OK, so our little risk calculator here is recommending that we would swab you for strep throat, and I agree with that."	Richardson 2019
	Risk justifies inaction	*"…there's a whole lot of people and this is your chance out of them of cancer…"*	Walker 2007
		"I think FRAX [is] useful because [it] can cut down on unnecessary investigations"	Voruganti 2015
	Improves understanding of risk	*"I think the biggest impact is that it changed the way I talked about what I was doing with patients, in that it made it a much more slick, neat package to describe the normal screening that you do for risk management. And so I felt it was easier to deliver some description of where they're at now. And from their point of view, I mean it's hard to know, but they seemed to understand that it was a multifactorial thing, rather than just being one of those single disease problems.... The thing that I don't really know, that I guess would be useful, is what they think when they walk out the door, what they actually understand of what I've said."*	Pieris 2009
		"With our Framingham tool you get…, a yellow bar. The average person's 10-year risk is…two little yellow bars, and your [the patient's] average risk is…maybe five little yellow bars'"	Voruganti 2015

		"I like this one [referring to the color bar].... I mean, everyone knows that red means danger, so if they're heading towards this one, it's a lot more visual, the impact...."	Peiris 2009
Autonomy	*Empowered by knowing personalised risk*	*"Anything that can be personalised or tailored, so you don't feel it's this off the shelf thing that you're being given, you know… you sit with your doctor and it's not just a case of giving out a leaflet, but actually let's have a look at your personal data"*	Porat 2019
		"I think the benefit is you're empowering yourself with information, you're possibly changing what you can change"	Keogh 2019, Patient comment
		"Anything that gets people to have a think about their health is a good thing, anything at all."	Walker 2007
Fear	*Direct fear of diagnosis*	*"Well I just kind of—a heart attack is a scariest thing to me. And I think it just kind of got me wondering and thinking and got me too nervous about the whole thing"*	Eaton 2009, Patient comment
		"And I thought, 'Wow! Patients coming with these tummy pains,' even if I'm thinking, 'She's got a 10% chance,' to actually think, 'Wow! Listen, you got a 38% chance of cancer.' General public are very—I would find that confronting and the general public have a shocking record at understanding risk."	Chiang 2015
	Disturbs normal practice	*"It's completely different to how I normally work, unsettling and confusing, I'm not quite sure where I am, to what extent I'm following a template and how I take history."*	Porat 2017
		"I personally don't think tools help me too much, interfere with my process of thought, would have done better without it, it was unnatural"	Porat 2017
	Removes focus and control of consultation	*"it might bring up a lot more other issues that they [patients] weren't originally aware of and the discussion might actually… be less directed"*	Voruganti 2015
		'Yeah. If I start saying to them, 'Oh god, your risk of having gastro-oesophageal cancer is one in three'y that seems way over the top, back pedalling from that, the consultation would've gone on for another 20 minutes or 30 minutes since I tried to explain to him, no. He doesn't have it'	Chiang 2015
	Lack of agency over risk	*The most commonly expressed concern was the potential for the risk assessment tool to miss something that was unknown and outside their control*	Kukafka 2015, Researcher Comment
	Medico-legal fear	*"Quite a few partners were worried about any medico-legal implications with*	Dikomitis 2015

		that. I think the one criticism was having the list of patient with PPVs. No one really liked that. That was worrying. We're worried about if patients knew that you had a list of them with the risk and you hadn't acted on it, what would be the implications? That was probably a point that put people off, really"	
		Some also felt that overriding a DDDSS would make them vulnerable to litigation	McParland 2019, Researcher Comment
Mistrust	*eRAT over-triggers*	*"Inevitably it will come up with a diagnosis of cancer. When the patient walks through the door . . . quite clearly, quickly, we can see this is not a serious diagnosis"*	McParland 2019
		"It gives information which, as it's blandly presented, you go, "How did you get that?..." I got a couple of people where I got a 20% number and you go, "Oh that's madness, that's not you," and often because it's based on single digit information…like a single blood pressure."	Peiris 2009
	eRAT misses things	*A consistent theme in our analysis was uncertainty among participants that all factors were accounted for in determining their breast cancer risk*	Kukafka 2015, Researcher comment
	eRAT not applicable to their patient	*In half of the patient visits, patients reported some piece of information, typically as a part of the history of present illness that raised a question for the coders of whether the tool was applicable to their clinical condition.*	Richardson 2019, Researcher comment
		they felt obliged to use the tool for the purpose of the study and their suspicion of cancer was unnaturally heightened	Chiang 2015
	Preference for diagnostic tests over risk scores	*"I would like to know if I have the gene mutation. So for me to say that being my mom had it [breast cancer] and take a pill and prevent it and not knowing if I have the gene doesn't make sense to me."*	Kukafka 2015
		"Colonoscopy was commonly referred to as the 'gold standard' for CRC screening. GPs were more likely to recommend colonoscopy if the patient wanted one or they had previously been recommended one by a gastroenterologist, even if the patient was at average risk."	Walker 2007, Researcher comment
	Preference for clinical judgement	*"… more experienced GPs were less likely to agree with the underlying concept and continued to maintain strong belief in their clinical intuition."*	Chiang 2015, Researcher comment
		"Patients don't follow flow charts, in my experience"	McParland 2019

		…that's just the tricky bit of saying [to the patient] that this is what's [sic] the recommendation but we don't always follow what recommendations are…	Walker 2007
	Patient preference for self knowledge	*"I think the model is incorrect because I know I am going to last longer than 10 years, because I do take care of myself"*	Saver 2014